MORVANE WERKZEUGMASCHINEN

MTC-420 CNC Turning Center

Operation and Maintenance Manual

Two-axis slant-bed CNC lathe with 12-station turret and MC-8T control

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Valid for	MTC-420 from serial number MTC420-23-0100
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Demo document. Morvane Werkzeugmaschinen and the MTC-420 are fictional and were created for the HealthyFac demo data.

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1 About This Manual

This manual describes the transport, installation, commissioning, operation, monitoring, maintenance, and decommissioning of the MTC-420 CNC turning center. It is the translation of the original instructions in German. If the two texts differ, the German text applies.

The manual is written for three target groups. **Operators** load workpieces, start programs, and do the daily checks. **Maintenance technicians** do the periodic maintenance and simple repairs; they need a mechanical or electrical qualification. **Morvane service** does the tasks that are marked as such, for example the spindle bearing replacement and the geometry check.

Keep this manual at the machine for its full service life. Give it to the next owner if you sell the machine.

Symbols and Signal Words

Signal word	Meaning
DANGER	A hazard that causes death or severe injury if you do not avoid it.
WARNING	A hazard that can cause death or severe injury.
CAUTION	A hazard that can cause a minor injury.
NOTICE	A risk of damage to the machine or the workpiece.

Related Documents

Document	Number	Content
Programming manual MC-8T	MC8T-PRG-EN	G codes, cycles, tool data, program management
Alarm manual MC-8T	MC8T-ALM-EN	Control alarms 10000 to 39999 (NC kernel, drives, PLC system)
Circuit diagram	MTC420-EP-24-0187	Wiring of this machine (in the electrical cabinet)
Hydraulic diagram	MTC420-HY-C	Hydraulic unit, chuck and tailstock circuits
Chip conveyor manual	CC2-OM-EN	Hinged belt conveyor CC-2

Revision History

Revision	Date	Change
A	2023-02	First edition.
B	2023-10	New vibration limits after field tests; alarm A111 added.
C	2025-03	Coolant concentration target changed to 6 to 8 %; maintenance interval of the hydraulic oil changed to 2,000 h.

2 Safety

Intended Use

The MTC-420 is intended for the turning, drilling, and threading of metal workpieces (steel, cast iron, aluminium, brass) in industrial use. The workpiece must fit within the working range of chapter 6 and must be clamped in the chuck, and supported by the tailstock when it is longer than 3 times its diameter.

Any other use is not intended. Foreseeable misuse includes: machining magnesium without a fire protection system, machining wood or plastics that produce fine dust, running with the guard door bridged, and using chucks that are not approved for the spindle speed.

DANGER - Rotating chuck. Never open the guard door or reach into the work area while the spindle turns. The door interlock stops the spindle, but a spindle that coasts down can still cause severe injury.

WARNING - Hydraulic clamping. The chuck clamps the workpiece with hydraulic pressure. If the hydraulic pressure falls below 25 bar, the clamping force is no longer guaranteed and the workpiece can be thrown out of the chuck.

WARNING - Chuck speed limit. Each chuck has a maximum speed that is engraved on it. The control limits the spindle speed to the value in the chuck data (parameter P2210). Enter the correct value after each chuck change.

Residual Risks

Hazard	Where	Protective measure
Workpiece or jaw thrown out	Work area	Guard door with polycarbonate window, rated for the maximum chuck energy; hydraulic pressure monitoring (A201).
Cuts from chips and tools	Work area, chip conveyor	Chip hook, gloves only when the spindle is off.
Skin irritation from coolant	Work area, coolant tank	Correct concentration, skin protection plan, regular coolant change.
Electric shock	Electrical cabinet	Cabinet lock, work only by qualified electricians with the main switch locked out. The DC link of the drives holds a voltage for 5 minutes after switch-off.
Hot surfaces	Spindle housing, hydraulic unit	Temperature up to 60 °C during operation. Let the parts cool down before maintenance.
Noise	Operator position	Emission sound pressure level 78 dB(A). Wear hearing protection for long shifts near open machines.

Safety Devices and Test Intervals

Device	Function	Test	Interval
Emergency stop buttons (2)	Stop all motion (stop category 1)	Press each button during a slow spindle run	Weekly

Device	Function	Test	Interval
Guard door interlock	Locks the door while the spindle turns; stops the spindle when opened in manual mode	Try to open the door at 50 rpm; the door must stay locked	Weekly
Enabling switch (handwheel)	Allows motion in manual mode with the door open	Release the switch during an axis motion	Monthly
Hydraulic pressure switch	Prevents the spindle start below 25 bar	Lower the pressure at the reducing valve during maintenance	Every 2,000 h
Polycarbonate window	Retains parts	Visual check for cracks, cloudiness, coolant damage. Replace after 5 years or after a hit.	Weekly / 5 years

Lockout Procedure

1. Stop the program and set the mode to off.
2. Switch off the main switch Q1 and lock it with your personal padlock.
3. Wait 5 minutes. Check with a voltmeter that the DC link of the drives is below 50 V.
4. Release the hydraulic pressure: open the chuck once with the foot pedal, then switch off the hydraulic unit.
5. Put a tag with your name and the date on the main switch.

3 Transport and Installation

Transport

The machine is delivered on a wooden skid with the X and Z axes locked by red transport brackets. Lift the machine only at the four lifting eyes with a lifting beam and four slings of at least 2 m length. Do not lift it at the turret or the spindle housing. The center of gravity is marked on the machine with a yellow label.

Item	Value
Transport weight (with skid)	4,650 kg
Transport dimensions (L x W x H)	3,150 x 1,900 x 2,050 mm
Lifting points	4 lifting eyes M30, on the machine base
Storage temperature	-10 to 50 °C, dry, protected from frost if coolant was filled

Foundation and Space

The machine needs a flat industrial concrete floor of at least 200 mm thickness without joints under the machine. The floor load is 2,100 kg/m² over the footprint. Keep a free space of 800 mm at the back for the electrical cabinet door and of 1,000 mm at the chip conveyor outlet.

Levelling

1. Remove the transport brackets of the X and Z axes (keep them for a later transport).
2. Place the 6 levelling pads under the levelling screws.
3. Put a precision level (0.02 mm/m) on the bed guideways, first along Z, then across.
4. Adjust the levelling screws until both directions are within 0.02 mm/m.
5. Tighten the lock nuts and check the level again after 1 week of operation.

Connections

Connection	Requirement
Electrical supply	400 V ±10 %, 3 phases + PE, 50 Hz, fuse 35 A gL, cable 5 x 10 mm². Earth leakage protection type B, 300 mA, if required by local rules.
Compressed air	6 bar, dry and oil-free, connection G 1/2. Used for the chuck air blow and the window cleaning.
Network (option MC8T-IOT)	Ethernet RJ45 at the cabinet roof; OPC UA port 4840, MQTT client to a plant broker.
Coolant	Fill 180 L of water-mixed coolant at 6 to 8 % concentration before the first start.
Hydraulic oil	Delivered filled (12 L). Check the level at the sight glass before the first start.

4 Commissioning

Morvane service or a person trained by Morvane does the first commissioning. Record the results in the commissioning report that is delivered with the machine.

Checklist Before the First Start

Check	Expected result
Transport brackets removed	All 4 red brackets removed and stored
Machine level	Within 0.02 mm/m in both directions
Phase sequence	Correct (the hydraulic pump turns in the arrow direction)
Hydraulic oil level	Between min and max at the sight glass
Central lubrication	Grease reservoir full; 3 manual cycles run without alarm A401
Coolant	Tank filled, concentration 6 to 8 %
Emergency stop and door interlock	Tested as in chapter 2
Chuck data	Maximum chuck speed entered in parameter P2210

First Start

1. Switch on the main switch Q1. The control starts in about 60 seconds.
2. Acknowledge the power-on message and start the hydraulic unit. The pressure must reach 35 bar within 10 seconds.
3. Reference the axes: mode manual, key REF, first X, then Z.
4. Run the warm-up program O9000 (10 minutes).
5. Machine the test workpiece of drawing MTC-TW-01 and measure it. Roundness must be within 0.003 mm and cylindricity within 0.008 mm over 150 mm.
6. Write the spindle temperature, the spindle vibration, and the hydraulic pressure after the warm-up into the commissioning report. These are the reference values for later comparisons.

> **Reference values of this unit (commissioning 2024-09-16):** spindle temperature 36 °C after O9000, spindle vibration 1.1 mm/s at 3,000 rpm, hydraulic pressure 35.5 bar, coolant concentration 7 %.

5 Machine Description

The MTC-420 is a two-axis CNC turning center with a 45-degree slant bed of cast iron. The slant bed lets the chips fall into the chip conveyor and gives the operator good access to the chuck.

Assembly	Description
Machine bed	Cast iron EN-GJL-300, ribbed, with roller linear guides size 35 for the X and Z axes.
Main spindle	Spindle nose A2-6, bore 66 mm, front and rear angular contact bearings with grease lubrication for life (re-greasing only during a bearing overhaul). Temperature sensor PT100 and vibration sensor at the front bearing.
Spindle drive	AC spindle motor 15 kW, poly-V belt 8PK1200, ratio 1:1.2.
Feed axes	Servo motors with absolute encoders, ball screws 32 x 10 mm, X stroke 230 mm, Z stroke 620 mm.
Turret	Servo turret with 12 stations, VDI 30, Hirth coupling with hydraulic clamping, bidirectional indexing.
Tailstock	Hydraulic quill, MT4, quill stroke 100 mm, thrust 3 to 12 kN, positioned by the Z slide.
Hydraulic unit	Gear pump 8 L/min, 12 L tank, pressure 35 bar, return filter 10 µm, pressure transmitter.
Central lubrication	Piston pump with 2 L grease reservoir, interval 30 min, pressure switch.
Coolant system	180 L tank, immersion pump 2 bar / 40 L/min, level sensor, temperature sensor, filter basket.
Chip conveyor	Hinged belt conveyor CC-2, outlet height 1,050 mm, runs with the coolant pump or on demand.
Electrical cabinet	Cooling by a filter fan (cabinet temperature monitored), main switch Q1, drives, PLC, power supply 24 V DC.
Control	Morvane MC-8T with 15-inch touch panel, handwheel, signal export option MC8T-IOT.

6 Technical Data

Unit Identification

The rating plate is on the left side of the electrical cabinet. This copy of the manual was delivered with the unit below.

Item	Value
Model	MTC-420
Serial number	MTC420-24-0187
Year of manufacture	2024
Control	Morvane MC-8T, software version 4.2
Commissioning date	2024-09-16
Options	MC8T-IOT (signal export), chip conveyor CC-2

Working Range and Axes

Parameter	Value
Swing over bed	560 mm
Maximum turning diameter	420 mm
Maximum turning length	600 mm
Bar capacity	52 mm
Stroke X / Z	230 / 620 mm
Rapid traverse X / Z	24 / 30 m/min
Feed force X / Z	6 / 8 kN
Positioning accuracy (ISO 230-2)	0.005 mm
Repeatability	0.002 mm

Spindle and Turret

Parameter	Value
Spindle speed range	30 to 4,500 rpm
Spindle motor power	15 kW (30 min), 11 kW (continuous)
Spindle torque	286 Nm (30 min)
Chuck size	8 inch (210 mm), hydraulic three-jaw
Turret stations	12, VDI 30
Indexing time, adjacent station	0.2 s

Supply and Environment

Parameter	Value
Supply voltage	400 V, 3 phases, 50 Hz, 35 A fuse
Connected load	28 kVA
Hydraulic system pressure	35 bar nominal
Coolant tank	180 L
Compressed air	6 bar, 150 L/min
Machine weight	4,300 kg
Footprint (L x W)	2,950 x 1,750 mm
Permitted ambient temperature	10 to 35 °C
Relative humidity	up to 75 %, no condensation
Emission sound pressure level	78 dB(A)

7 Operation

Operator Panel

Key or element	Function
Cycle Start	Starts or continues the active program.
Feed Hold	Stops the axis feed; the spindle keeps turning (message S003).
Mode switch	Selects auto, manual, or off (see 7.2).
Spindle override	50 to 120 % of the programmed speed.
Feed override	0 to 150 % of the programmed feed.
Door key	Unlocks the guard door when the spindle is stopped.
Coolant key	Switches the coolant pump on and off in manual mode.
Emergency stop	Stops all motion; release by turning the button.

Operating Modes

The mode switch sets the spindle mode. The control reports the active mode as the signal **spindle.mode**.

Mode	Meaning
auto	The control runs the active program. The guard door must be closed and locked.
manual	Set-up mode. The operator moves the axes with the handwheel. With the door open, the spindle speed is limited to 50 rpm and the axis speed to 2 m/min.
off	The spindle drive is disabled. The control, the coolant pump, and the hydraulic unit can stay on.

Start of a Shift

1. Check the coolant level and the hydraulic oil level (sight glasses).
2. Switch on the machine and start the hydraulic unit.
3. Reference the axes if the control asks for it (after a power failure).
4. Run the warm-up program O9000. It runs the spindle for 10 minutes in steps from 500 to 3,500 rpm.
5. Check the spindle temperature on the diagnosis screen. After the warm-up, 30 to 45 °C is typical.

> **NOTICE.** A cold spindle that runs at full speed at once wears its bearings faster and can show a short temperature alarm.

End of a Shift

1. Finish the program and move the turret to the home position.
2. Remove chips from the work area and the turret with a brush.
3. Close the coolant nozzles' valves so that the coolant does not drain into the work area.
4. Set the mode to off. Leave the main switch on if the machine is used in the next shift.

8 Monitored Signals and Operating Limits

The table lists every signal that the MC-8T control reports. The normal range is the range for continuous operation. A value in the warning range is allowed for a short time but needs attention. A value at the alarm limit stops the machine.

Signal	Signal key	Unit	Normal range	Warning	Alarm (stop)
Spindle bearing temperature	spindle.temperature	°C	15 to 60	above 65	above 70 (A101)
Spindle speed	spindle.speed	rpm	0 to 4,500	-	-
Spindle load	spindle.load	%	0 to 100	100 to 120 (max. 30 min)	above 120 for 10 s (A102)
Spindle vibration (RMS velocity)	spindle.vibration	mm/s	0 to 2.8	above 4.5 (A110)	above 7.1 (A111)
Hydraulic pressure	hydraulic.pressure	bar	30 to 40	below 28	below 25 (A201)
Hydraulic oil temperature	hydraulic.oil_temperature	°C	20 to 50	above 55 (A203)	above 65 (A204)
Coolant level	coolant.level	%	30 to 100	below 25 (A305)	below 10 (A306)
Coolant temperature	coolant.temperature	°C	10 to 35	above 35 (A307)	-
Lubrication pressure	lubrication.pressure	bar	15 to 25 (during a cycle)	-	no pressure in 3 cycles (A401)
Cabinet temperature	cabinet.temperature	°C	15 to 40	above 45 (A801)	above 50 (A802)
Guard door open	door.open	true / false	-	-	open in auto mode (A501)
Spindle mode	spindle.mode	auto, manual, off	-	-	-
Operating hours	machine.operating_hours	h	-	-	-

Spindle temperature. The sensor sits at the front spindle bearing. A rise of more than 5 °C within 15 minutes at a constant speed is an early sign of a bearing or lubrication problem, also when the value is still in the normal range. Compare the value after the warm-up with the reference value of the commissioning report.

Vibration. The vibration limits follow ISO 10816-3 for machines of group 2 on a rigid foundation. Measure on the spindle housing in the radial direction. A new machine shows 0.8 to 1.5 mm/s. An unbalanced chuck

> or workpiece is the most common cause of a high value. A slow rise over months points to bearing wear.

Signal Export (Option MC8T-IOT)

The control publishes the signals every 5 seconds to an MQTT broker or offers them as OPC UA variables. The MQTT payload is a JSON object with the sample time and one property per signal key. Status codes are sent as a list of active codes. Example:

```
{ "sampledAt": "2025-04-02T08:15:05Z",
  "metrics": { "spindle.temperature": 41.2, "spindle.speed": 2800,
               "spindle.load": 64, "spindle.vibration": 1.3, "door.open": false,
               "spindle.mode": "auto", "coolant.level": 71 },
  "statuses": ["S002"] }
```

9 Alarms and Messages

The control shows each alarm with its code on the screen and stores it in the alarm history (last 1,000 entries). An error stops the program and the spindle. A warning lets the current program finish. An info message needs no action. Control alarms with five digits (for example 21612) are described in the alarm manual MC8T-ALM-EN.

Spindle (100)

Code	Type	Text	Cause and action
A101	Error	Spindle overtemperature	The spindle bearing temperature is above 70 °C. Let the spindle cool down to below 45 °C. Check the belt tension and the spindle cooling fins. If the alarm comes back within 1 week, have the spindle bearings inspected by Morvane service.
A102	Error	Spindle overload	The spindle load was above 120 % for more than 10 s. Reduce the depth of cut or the feed rate. Check the insert for wear; a blunt insert raises the load by 20 to 40 %.
A103	Error	Spindle speed deviation	The actual speed differs by more than 10 % from the set speed for 2 s. Check the belt for slip and the spindle encoder cable.
A104	Error	Spindle drive fault	The spindle drive reported a fault. Read the drive code on the diagnosis screen (alarm manual MC8T-ALM-EN).
A110	Warning	Spindle vibration high	The vibration is above 4.5 mm/s. Check the balance of the chuck and the workpiece and the clamping of long workpieces. Plan a vibration analysis if the value stays high.
A111	Error	Spindle vibration too high	The vibration is above 7.1 mm/s. The spindle stops. Do not restart before the cause is found: loose chuck, damaged bearing, loose belt pulley.
A112	Warning	Spindle temperature rising fast	The temperature rose by more than 5 °C within 15 minutes at a constant speed. Reduce the speed and watch the trend. Check the lubrication state of the bearings at the next stop.

Hydraulics (200)

Code	Type	Text	Cause and action
A201	Error	Hydraulic pressure low	The hydraulic pressure is below 25 bar. The chuck clamping is not safe; the spindle cannot start. Check the oil level, the pressure setting (35 bar), and the hoses for leaks.
A202	Warning	Hydraulic return filter clogged	The filter indicator switched. Replace the return filter element MV-420-2210 within 1 week.
A203	Warning	Hydraulic oil temperature high	The oil temperature is above 55 °C. Check the oil level and the cooler fins of the hydraulic unit. A hot ambient raises the oil temperature.

Code	Type	Text	Cause and action
A204	Error	Hydraulic oil overtemperature	The oil temperature is above 65 °C. The hydraulic unit stops; the chuck stays clamped by the check valve. Let the oil cool down and find the cause before the restart.
A205	Warning	Hydraulic oil level low	The level switch reports a low level. Add oil ISO VG 32 and look for leaks.

Coolant and chips (300)

Code	Type	Text	Cause and action
A301	Error	Coolant pump motor overload	The motor protection of the coolant pump tripped. Check the pump for chips and the filter basket.
A302	Error	Chip conveyor overload	The conveyor stopped because of a jam. Run the conveyor backwards for 5 s, remove the jam, and restart.
A305	Warning	Coolant level low	The coolant level is below 25 %. Add coolant at a concentration of 6 to 8 % before the end of the shift.
A306	Error	Coolant level critical	The coolant level is below 10 %. The coolant pump stops to protect itself from dry running. Fill the tank and acknowledge the alarm.
A307	Warning	Coolant temperature high	The coolant is above 35 °C. Thermal growth of the workpiece can affect the size. Check the tank level; consider a coolant chiller in summer.

Lubrication (400)

Code	Type	Text	Cause and action
A401	Error	Lubrication fault	The central lubrication pump built no pressure in 3 cycles in a row. Check the grease reservoir and the pump. Do not run the axes without lubrication for more than 1 hour.
A402	Warning	Grease reservoir low	The grease reservoir is below 20 %. Refill grease NLGI class 00.

Safety (500)

Code	Type	Text	Cause and action
A501	Error	Door open in automatic mode	The guard door was opened while a program ran in auto mode. Close the door and restart the program from the last block.
A502	Error	Door lock fault	The door lock did not reach its position. Check the lock and the door alignment.
A503	Error	Emergency stop active	An emergency stop button is pressed. Release it and acknowledge.
A504	Error	Chuck not clamped	The clamp sensor reports no clamped state at the program start. Check the jaw stroke and the sensor position.

Axes and turret (600 and 700)

Code	Type	Text	Cause and action
A601	Error	X axis following error	The X axis deviates more than allowed from its path. Check for a collision and for chips on the guide covers.
A602	Error	Z axis following error	As A601 for the Z axis.
A603	Error	Software limit switch	The program moves an axis outside its stroke. Correct the program or the work offset.
A604	Warning	Axis reference lost	The absolute position was lost (encoder battery). Replace the encoder battery and reference the axis.
A701	Error	Turret not clamped	The turret did not reach its clamped position after indexing. Check the hydraulic pressure and the turret for chips.
A702	Error	Turret indexing timeout	The indexing took longer than 2 s. Check the turret motor and the clamping.

Electrical cabinet (800)

Code	Type	Text	Cause and action
A801	Warning	Cabinet temperature high	The cabinet temperature is above 45 °C. Clean or replace the filter mat. Check the cabinet fan.
A802	Error	Cabinet overtemperature	The cabinet temperature is above 50 °C. The drives stop to protect the electronics. Find the cause before the restart.
A803	Warning	24 V supply low	The 24 V DC supply is below 22 V. Check the power supply and the load.

Maintenance and states (900 and S)

Code	Type	Text	Cause and action
M900	Warning	Maintenance due	The operating hours reached the next maintenance interval of chapter 10. Do the listed tasks and reset the counter on the maintenance screen.
M901	Info	Encoder battery due	Replace the encoder batteries within 4 weeks (every 2 years).
S001	Info	Ready	The machine is ready for a program start.
S002	Info	Cycle running	A program runs in auto mode.
S003	Info	Feed hold	The operator stopped the axis feed. Press Cycle Start to continue.
S004	Info	Warm-up running	The warm-up program O9000 runs.

10 Maintenance

The intervals count operating hours (signal machine.operating_hours) or calendar time, whichever comes first. The control shows message M900 when a 500-hour interval is reached. Record each task in the maintenance log (section 10.4). Lock out the machine (chapter 2) before every task inside the work area or the cabinet.

Maintenance Schedule

Interval	Task	Time	Who
Each shift	Check the coolant level and the hydraulic oil level. Remove chips from the work area and the turret. Run the warm-up program O9000.	10 min	Operator
Weekly	Measure the coolant concentration with a refractometer (target 6 to 8 %). Clean the chip conveyor filter basket. Check the guard door window for cracks. Check the grease level of the central lubrication. Test the emergency stop buttons and the door interlock.	30 min	Operator
500 h or 3 months	Clean the coolant tank and change the coolant. Check the poly-V belt tension. Clean the electrical cabinet filter mat. Check the chuck jaws for wear and grease the chuck. Check the wipers of the guide covers.	2 h	Maintenance technician
2,000 h or 1 year	Change the hydraulic oil (12 L, ISO VG 32) and the return filter. Check the backlash of the X and Z axes (limit 0.008 mm). Check the turret clamping and the coolant nozzles. Replace the cabinet filter mat. Test the hydraulic pressure switch.	4 h	Maintenance technician
2 years	Replace the encoder batteries of the X and Z axes (with the control switched on).	20 min	Maintenance technician
8,000 h or 4 years	Inspect the spindle bearings (vibration analysis and run-out check, limit 0.003 mm). Replace the poly-V belt. Check the ball screws and the linear guides. Check the geometry with a test workpiece.	1 day	Morvane service
5 years	Replace the polycarbonate window of the guard door.	1 h	Maintenance technician

Procedures

Check the Poly-V Belt Tension

1. Lock out the machine and remove the rear cover of the spindle drive.
2. Press the belt in the middle between the pulleys with a spring scale at 50 N.
3. The deflection must be 4 to 5 mm. Adjust the motor plate with the two tension screws if needed.
4. A belt with cracks, shiny flanks, or missing ribs must be replaced (MV-420-1150).

Change the Hydraulic Oil

1. Run the machine for 15 minutes so that the oil is warm and the dirt is in suspension.
2. Lock out the machine. Place a 15 L container under the drain valve of the hydraulic tank.
3. Drain the oil and replace the return filter element (MV-420-2210).
4. Fill 12 L of fresh oil ISO VG 32 through the filler filter.

5. Start the hydraulic unit, clamp and unclamp the chuck 10 times, and check the level again.

Change the Coolant

1. Pump the old coolant out with a coolant vacuum unit. Dispose of it as hazardous waste (code 12 01 09*).
2. Remove the chips and the sludge from the tank and clean it with a system cleaner.
3. Rinse the tank and the pipes with water.
4. Fill 180 L of new coolant. Mix the concentrate into the water (never water into the concentrate) to 7 %.

Grease the Chuck

Open the jaws completely. Press 2 strokes of chuck grease into each grease nipple. Clamp and unclamp 5 times without a workpiece. A chuck without grease loses up to 50 % of its clamping force.

Lubricants and Fluids

Point	Lubricant	Quantity
Hydraulic unit	Hydraulic oil HLP, ISO VG 32	12 L
Central lubrication	Grease NLGI class 00	2 L reservoir
Chuck	Chuck grease (high pressure)	2 strokes per nipple
Spindle bearings	Lubricated for life	-
Coolant	Water-mixed cutting fluid, mineral oil free, 6 to 8 %	180 L

Maintenance Log

Copy this page or use the maintenance screen of the control. The last entries of this unit are listed as an example.

Date	Hours	Task	Remarks	Name
2024-09-16	0	Commissioning	Reference values recorded	Morvane service
2024-12-12	512	500 h maintenance	Belt tension corrected (6 mm before)	J. Weber
2025-03-20	1,040	500 h maintenance	Coolant changed early: smell	J. Weber
2025-06-30	1,580	500 h maintenance	Chuck jaws replaced (MV-420-4012)	A. Novak

11 Troubleshooting

Find the symptom in the table and work through the causes in the given order. If the problem stays, call Morvane service with the alarm history and the serial number.

Symptom	Possible cause	Action
Spindle temperature rises slowly over several days	Bearing preload changed, belt too tight, cooling fins dirty	Check the belt tension. Clean the fins. Compare the vibration with the reference value. Call Morvane service for a bearing inspection.
Spindle temperature high only in the afternoon	Hot ambient, coolant temperature high	Check the hall temperature and the coolant temperature (A307).
Rough surface on all workpieces	Worn insert, high spindle vibration, loose turret	Change the insert. Read the spindle vibration. Check the turret clamping.
Chatter marks on long workpieces	No tailstock support, wrong cutting data	Use the tailstock above a length of 3 times the diameter. Reduce the depth of cut or change the speed by 10 %.
Diameter changes during the shift	Thermal growth of the spindle, coolant temperature changes	Run the warm-up program. Keep the coolant temperature stable.
Hydraulic pressure drops during the shift	Low oil level, internal leak in a valve, hot oil	Check the oil level and the oil temperature (max. 55 °C). Check the valves for leaks.
Chuck clamps slowly	Low pressure, dirty chuck, missing grease	Check the pressure. Clean and grease the chuck.
Coolant smells or foams	Bacterial growth, wrong concentration, oil in the coolant	Measure the concentration. Skim off tramp oil. Change the coolant if the smell stays.
Alarm A401 after a long stop	Air in the lubrication lines	Run 5 manual lubrication cycles from the maintenance screen.
Turret does not index (A702)	Chips in the Hirth coupling, low hydraulic pressure	Clean the turret. Check the pressure.
Axis position wrong after power-on (A604)	Encoder battery empty	Replace the battery and reference the axis.
Cabinet alarm A801 in summer	Dirty filter mat, fan fault	Replace the filter mat. Check the fan. Keep the cabinet door closed.
Chip conveyor stops (A302)	Long chips wrap around the belt	Use a chip breaker geometry. Run the conveyor backwards and remove the chips.

12 Electrical Data

The circuit diagram of this machine is in the pocket of the electrical cabinet door. Only qualified electricians may work in the cabinet.

Main Components

Item	Component	Data
Q1	Main switch	63 A, lockable
F1	Main fuses of the drives	3 x 35 A gL
T1	Control transformer	400 / 230 V, 1 kVA
G1	Power supply 24 V DC	20 A
U1	Spindle drive	18.5 kW converter
U2, U3	Servo drives X and Z	4.5 kW each
U4	Turret drive	1.5 kW
M5	Hydraulic pump motor	1.5 kW, 400 V
M6	Coolant pump motor	0.75 kW, 400 V
M7	Chip conveyor motor	0.25 kW, 400 V
E1	Cabinet filter fan	230 V, 160 m³/h

Terminal Strip X20 (Signal Interface)

Terminal	Signal	Type
X20.1	Machine ready	Output 24 V DC
X20.2	Alarm active	Output 24 V DC
X20.3	Cycle running	Output 24 V DC
X20.4	Maintenance due	Output 24 V DC
X20.5	External stop	Input, normally closed
X20.6	Bar feeder ready	Input 24 V DC
X20.7 / X20.8	0 V / 24 V DC	Supply

13 Spare Parts

Order spare parts from Morvane service. State the serial number of the machine and the part number.

Spindle

Part number	Description	Quantity
MV-420-1100	Spindle unit, complete (exchange part)	1
MV-420-1110	Spindle bearing set, front and rear	1
MV-420-1120	PT100 temperature sensor, spindle	1
MV-420-1130	Spindle encoder	1
MV-420-1150	Poly-V belt 8PK1200	1
MV-420-1160	Belt pulley, motor side	1
MV-420-6005	Vibration sensor, spindle housing	1

Chuck and tailstock

Part number	Description	Quantity
MV-420-4010	Hydraulic chuck 8 inch, complete	1
MV-420-4012	Chuck jaw set, hard, 3 pieces	1
MV-420-4014	Chuck jaw set, soft, 3 pieces	1
MV-420-4020	Clamping cylinder seal kit	1
MV-420-4030	Live centre MT4	1
MV-420-4040	Tailstock quill seal kit	1

Hydraulics

Part number	Description	Quantity
MV-420-2200	Gear pump 8 L/min	1
MV-420-2210	Return filter element 10 µm	1
MV-420-2220	Pressure transmitter 0-100 bar	1
MV-420-2230	Pressure reducing valve	1
MV-420-2240	Level switch, hydraulic tank	1
MV-420-2250	Hose set, chuck circuit	1

Coolant, chips, and lubrication

Part number	Description	Quantity
MV-420-3010	Coolant pump 2 bar / 40 L/min	1

Part number	Description	Quantity
MV-420-3020	Level sensor, coolant tank	1
MV-420-3025	Filter basket, coolant tank	1
MV-420-3030	Filter mat, electrical cabinet	1
MV-420-3110	Lubrication pump, complete	1
MV-420-3120	Metering valve set (12 pieces)	1
CC2-0410	Hinged belt, chip conveyor	1

Axes and turret

Part number	Description	Quantity
MV-420-5010	Ball screw X, 32 x 10	1
MV-420-5020	Ball screw Z, 32 x 10	1
MV-420-5030	Wiper set, guide covers	1
MV-420-5040	Encoder battery pack	2
MV-420-7010	Turret clamping seal kit	1
MV-420-7020	Coolant nozzle set, turret	1

Safety and cabinet

Part number	Description	Quantity
MV-420-8010	Polycarbonate window, guard door	1
MV-420-8020	Door lock with interlock switch	1
MV-420-8030	Emergency stop button	1
MV-420-8040	Cabinet filter fan	1
MV-420-8050	Power supply 24 V DC, 20 A	1

Morvane Service: service@morvane.example, phone +49 000 000000.

14 Decommissioning and Disposal

For a stop of more than 3 months, clean the machine, drain the coolant, oil the bare surfaces, and move the axes to the middle of their strokes. Run the warm-up program and 5 lubrication cycles before the restart.

For the final decommissioning, drain all fluids and dispose of them by type: hydraulic oil as waste oil, coolant as hazardous waste, grease with the oil waste. Remove the encoder batteries and dispose of them separately. The machine body is cast iron and steel and can be recycled as scrap metal. The electrical components are electronic waste.

Tightening Torques

Thread	Strength class 8.8	Strength class 10.9
M6	10 Nm	14 Nm
M8	25 Nm	35 Nm
M10	49 Nm	69 Nm
M12	86 Nm	120 Nm
M16	210 Nm	295 Nm

EC Declaration of Conformity (Extract)

The manufacturer declares that the machine MTC-420 CNC turning center, serial number MTC420-24-0187, complies with the Machinery Directive 2006/42/EC and the EMC Directive 2014/30/EU. Applied standards include EN ISO 23125 (safety of turning machines), EN 60204-1, and EN ISO 12100. The full declaration is delivered as a separate document.